SIKSHANA FOUNDATION

Class 10 Science: Light, Lenses and Mirrors

Sikshanapedia Editors

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Contents

1. Reflection and plane mirrors 3
2. Spherical mirrors and ray diagrams 9
3. The mirror formula 16
4. Refraction and lenses 22
5. The lens formula and power 29
6. The human eye 35
7. Practice papers 42

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CHAPTER 1

Reflection and plane mirrors

A revision book for the SSLC optics unit, built from the most-viewed Sikshanapedia lessons. Ray diagrams drawn step by step, the mirror and lens formulas with worked examples, and the sign convention explained once, properly. Ends with three full practice papers.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred villages is data, and data is how we learn things nobody has learned

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The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

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Spherical mirrors and ray diagrams

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CHAPTER 3

The mirror formula

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Refraction and lenses

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CHAPTER 5

The lens formula and power

A lens does one thing: it bends light. A convex lens bends rays together to a point we call the focus. A concave lens spreads them apart. Everything else, the spectacles, the microscope, the camera, the telescope on the roof of the library, is a careful arrangement of this one idea.

Before you look for anything in the night sky, give your eyes twenty minutes. In that time the pupils widen and a chemical in the retina rebuilds itself, and stars that were invisible when you stepped outside will simply appear, as if someone had switched them on. Keep your phone in your pocket; one bright screen undoes the whole twenty minutes.

A shadow is only ever as long as the angle of the light allows. At sunrise and sunset the Sun is low and shadows stretch across the whole field. At noon the Sun is high and the shadow shrinks to a small patch at your feet. On two days of the year, at places between the tropics, it shrinks to nothing at all.

Try this: hold your arm straight out and make a fist. The width of your fist against the sky is about ten degrees. Three fists stacked from the horizon is thirty degrees, which on a December evening in Karnataka is roughly where you will find Jupiter. Astronomers have used this trick for thousands of years, and it still works.

Record what you see, even when it seems like nothing. Date, time, where you stood, how clear the sky was, what you noticed. A single observation is a story; a hundred observations from a hundred

villages is data, and data is how we learn things nobody has learned before.

The light from a lamp falls off quickly as you walk away from it. Double the distance and the light on your page is a quarter as bright. This is the inverse square law, and it is why a candle across the room is dim while the Sun, which is a hundred and fifty million kilometres away, can still burn your skin.

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CHAPTER 6

The human eye

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CHAPTER 7

Practice papers

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